

# GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of canopy height per plot (esy4)

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**NOTE:** To compare different models, you only have to change the models in the section ‘Load models’

## Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(DHARMA)
library(emmeans)
```

## Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_esy4.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    fertilized = "f",
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
mutate(
  esy4 = fct_relevel(esy4, "R", "R22", "R1A"),
  eco.id = factor(eco.id)
) %>%
rename(y = cwm.abu.height) %>%
filter(y < 1) # see section Outliers: Exclude site N_DAM (more or less only the tall grass Arrhenathera)
```

## Load data

## Statistics

### Data exploration

#### Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 0.4776193 0.4676948 0.4577703
```

```
median(sites$y)
```

```
## [1] 0.47
```

```
sd(sites$y)
```

```
## [1] 0.1254282
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
## 5% 95%
## 0.27 0.68
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id      n
##   <fct> <int>
## 1 654      202
## 2 664      199
## 3 686      215
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord> <int>
## 1 positive     102
## 2 restored     396
## 3 negative     118
```

```
sites %>% count(esy4)
```

```
## # A tibble: 3 x 2
##   esy4      n
##   <fct> <int>
## 1 R      327
## 2 R22    208
## 3 R1A     81
```

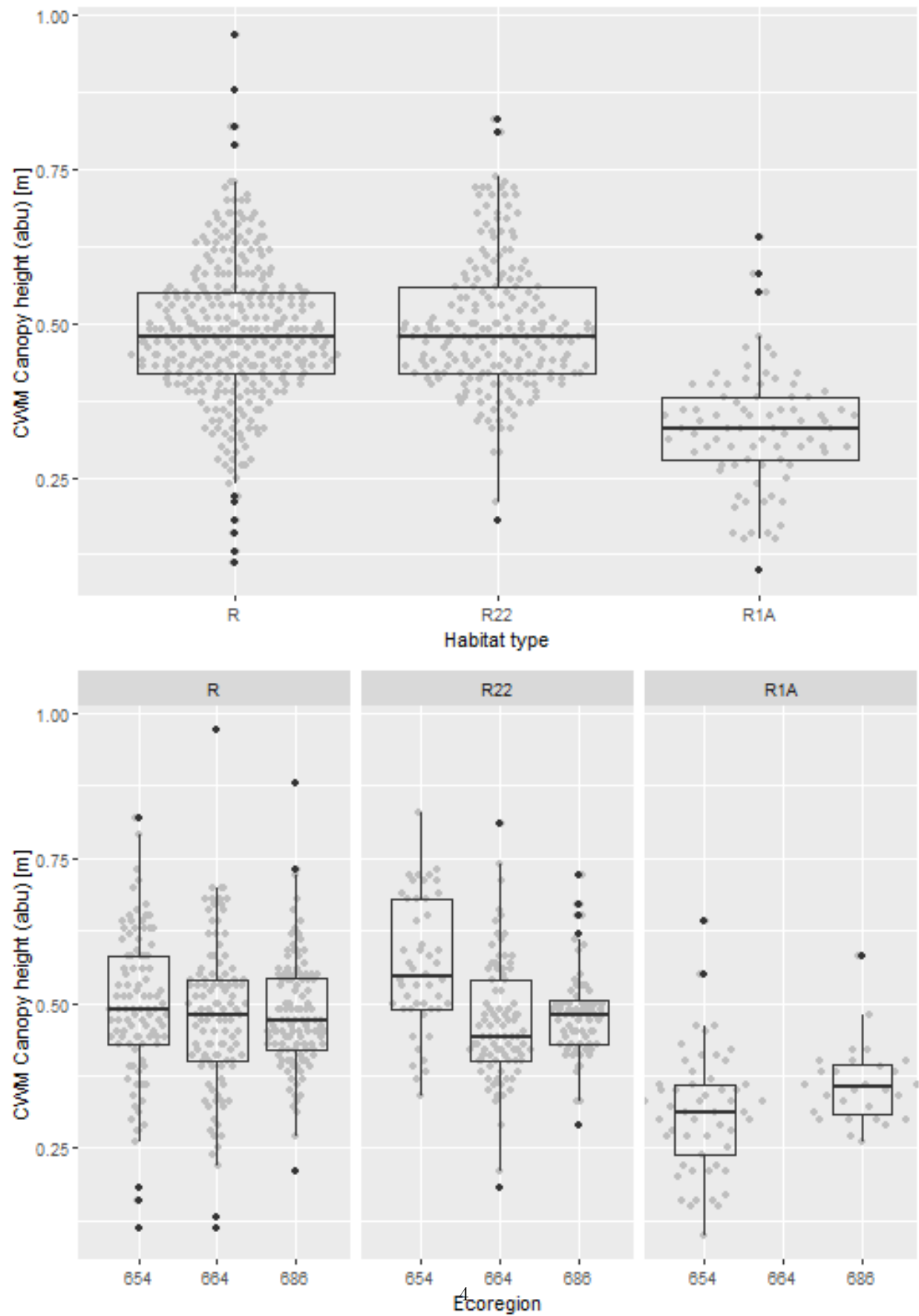
```
sites %>% count(esy4, eco.id)
```

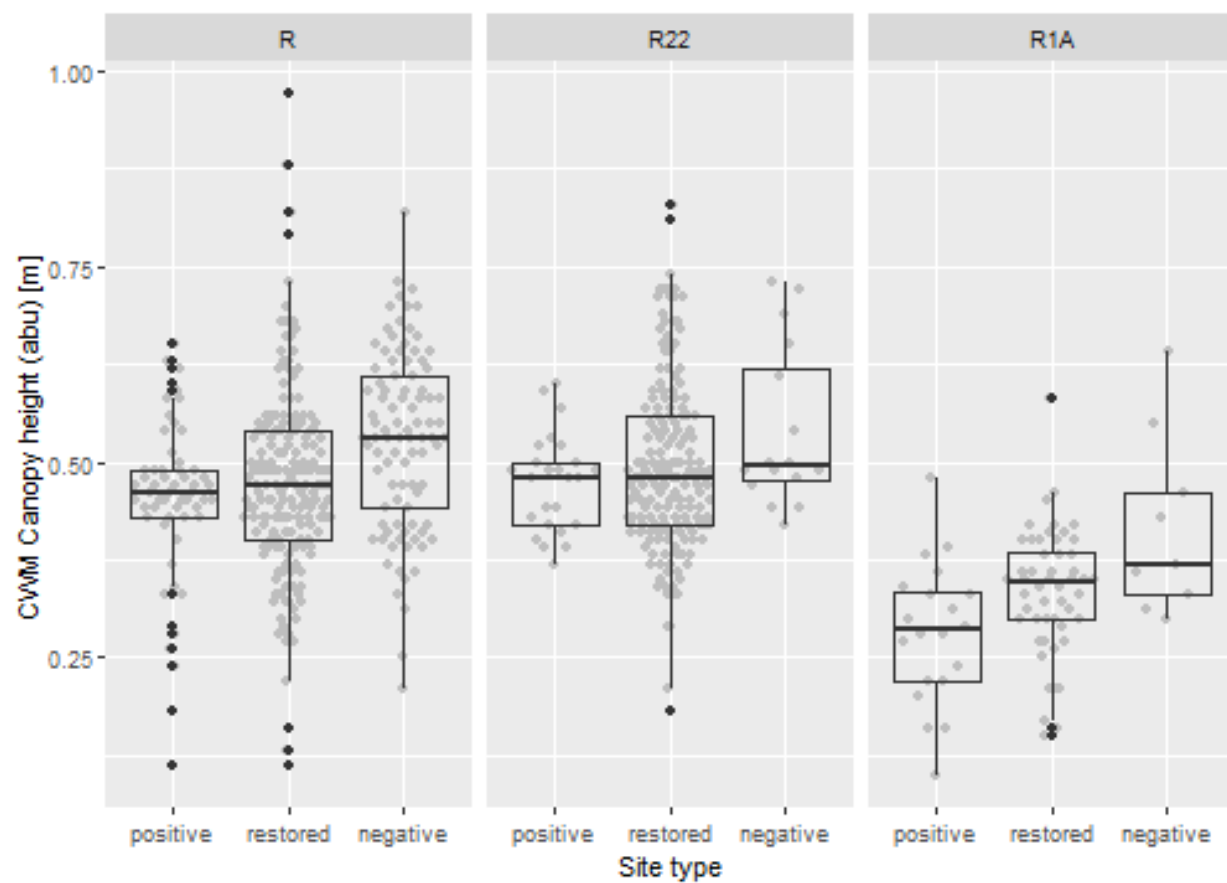
```
## # A tibble: 8 x 3
##   esy4 eco.id      n
##   <fct> <fct> <int>
## 1 R     654     101
## 2 R     664     110
## 3 R     686     116
## 4 R22   654      48
## 5 R22   664      89
## 6 R22   686      71
## 7 R1A   654      53
## 8 R1A   686      28
```

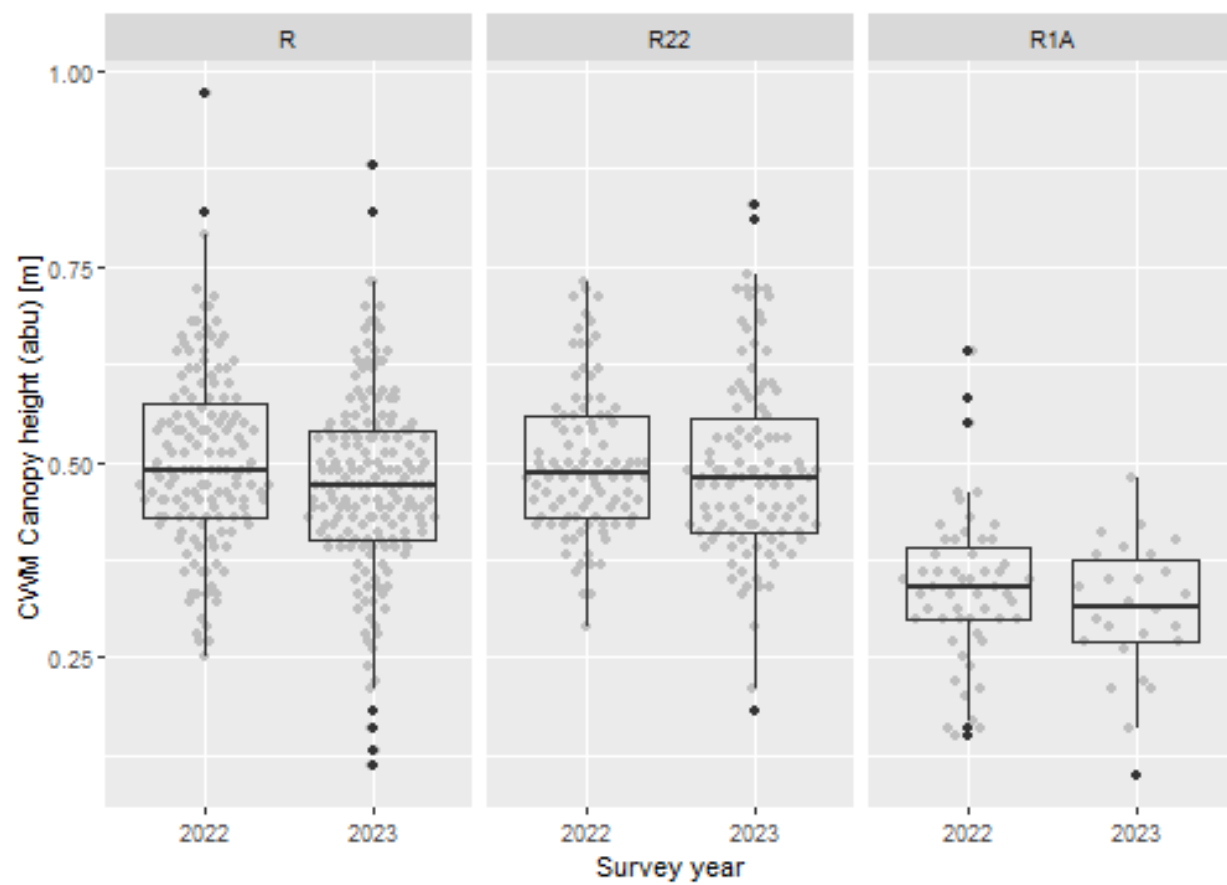
```
sites %>% count(esy4, site.type)
```

```
## # A tibble: 9 x 3
##   esy4 site.type      n
##   <fct> <ord> <int>
## 1 R     positive     57
## 2 R     restored    177
## 3 R     negative     93
## 4 R22   positive     25
## 5 R22   restored    167
## 6 R22   negative     16
## 7 R1A   positive     20
## 8 R1A   restored     52
## 9 R1A   negative      9
```

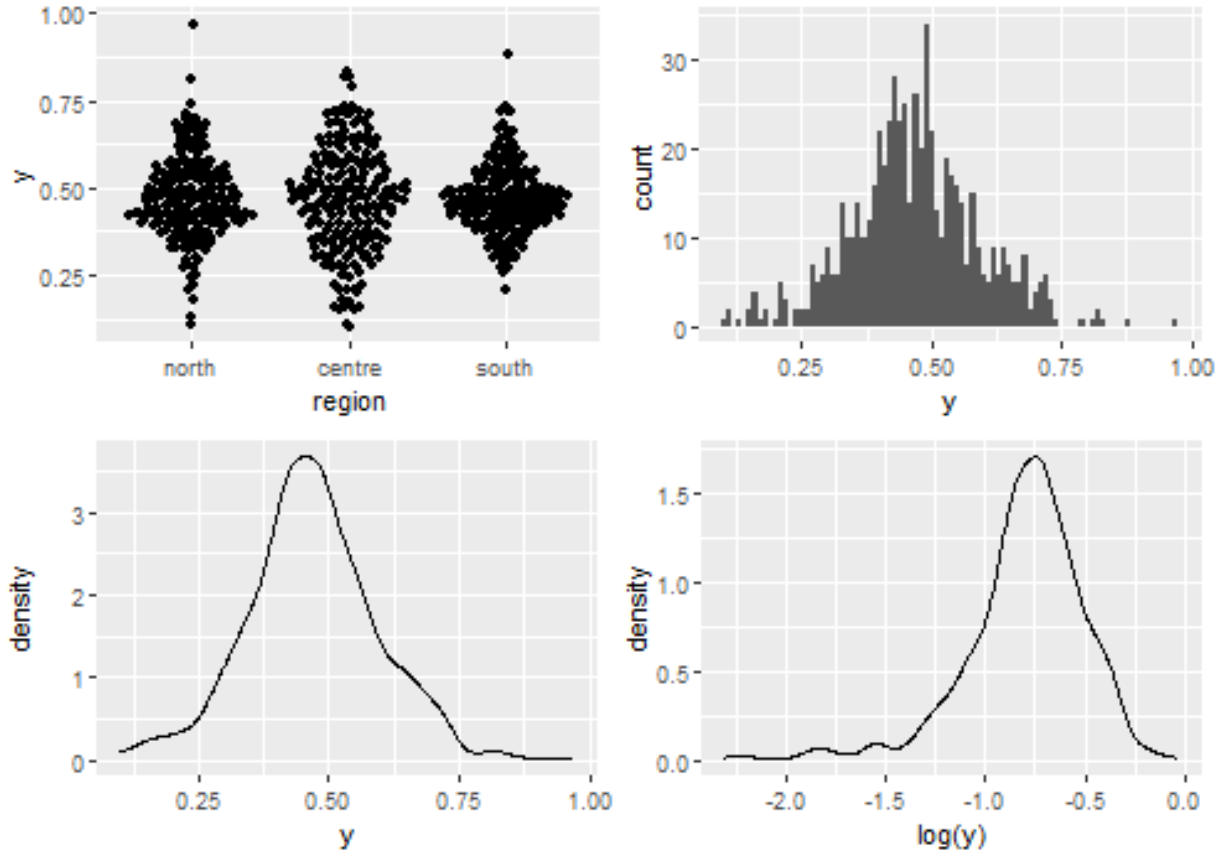
Graphs of raw data (Step 2, 6, 7)







Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude  $r > 0.7$  Dormann et al. 2013 *Ecography* DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%  
#   select(where(is.numeric), -y, -starts_with("cum.)) %>%  
#   GGally::ggpairs(  
#     lower = list(continuous = "smooth_loess")  
#   ) +  
#   theme(strip.text = element_text(size = 7))  
  
# -> no continuous variables
```

## Models

**NOTE:** Only here you have to modify the script to compare other models

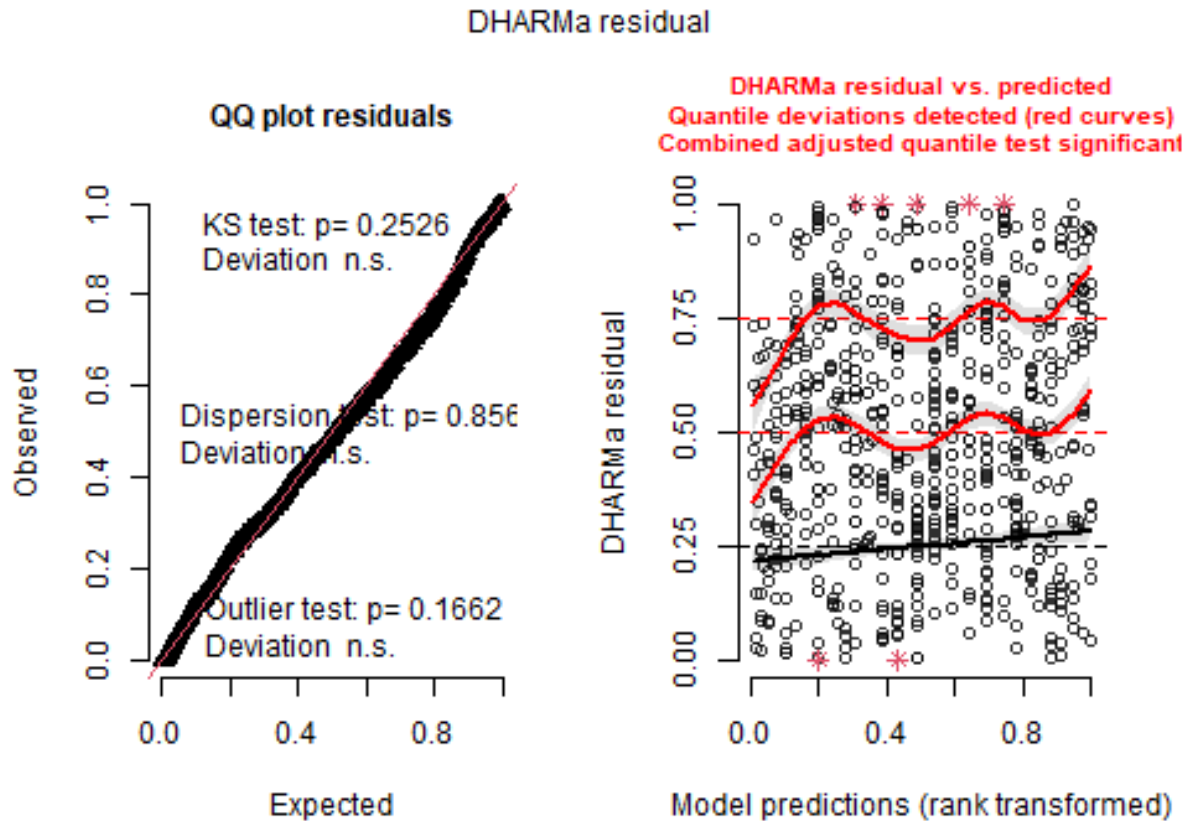
```
load(file = here("outputs", "models", "model_height_esy4_2.Rdata"))  
load(file = here("outputs", "models", "model_height_esy4_3.Rdata"))  
m_1 <- m2  
m_2 <- m3  
  
m_1@call  
## lmer(formula = y ~ esy4 * site.type + eco.id + obs.year + (1 |  
##   id.site), data = sites, REML = FALSE)
```

```
m_2@call
## lmer(formula = y ~ esy4 * (site.type + eco.id) + obs.year + hydrology +
##       (1 | id.site), data = sites, REML = FALSE)
```

## Model check

### DHARMA

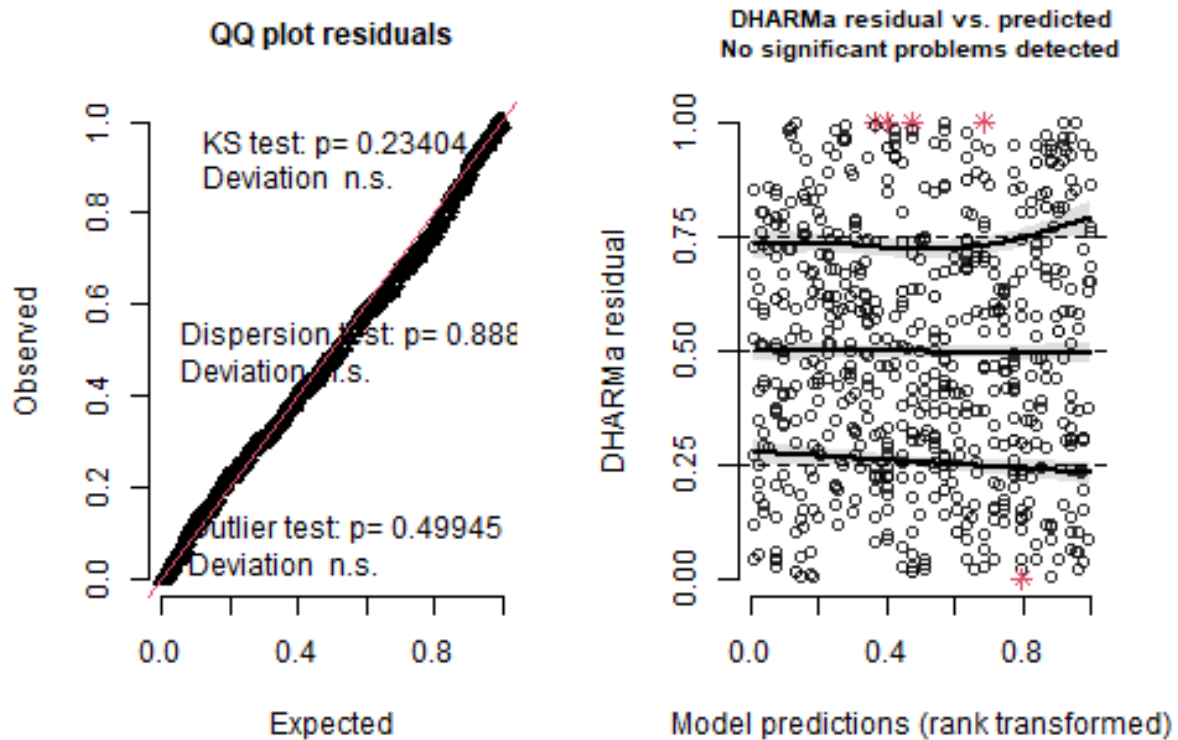
```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



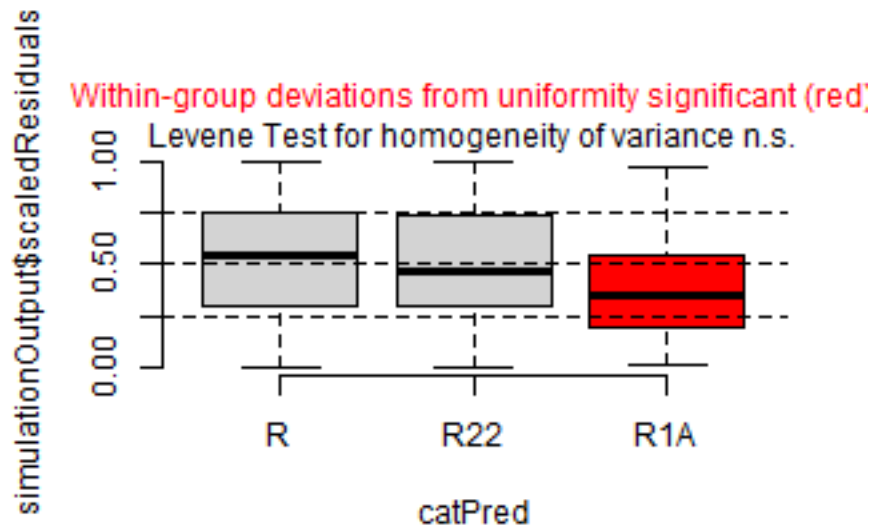
```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



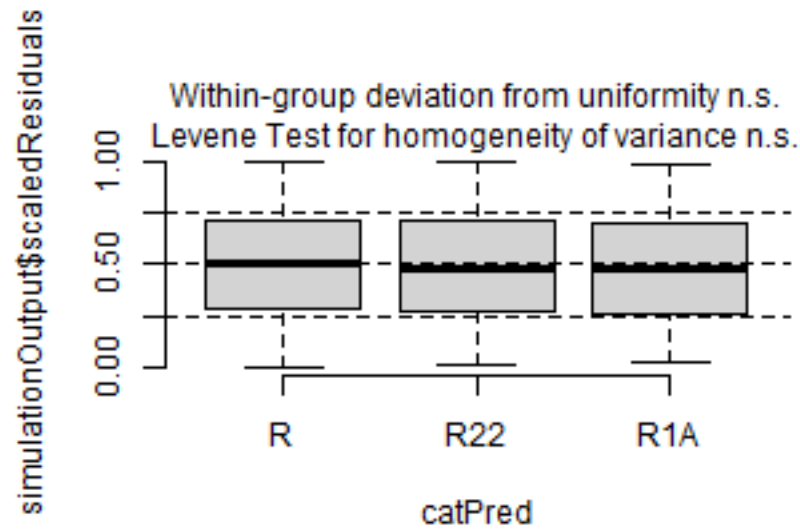
## DHARMa residual



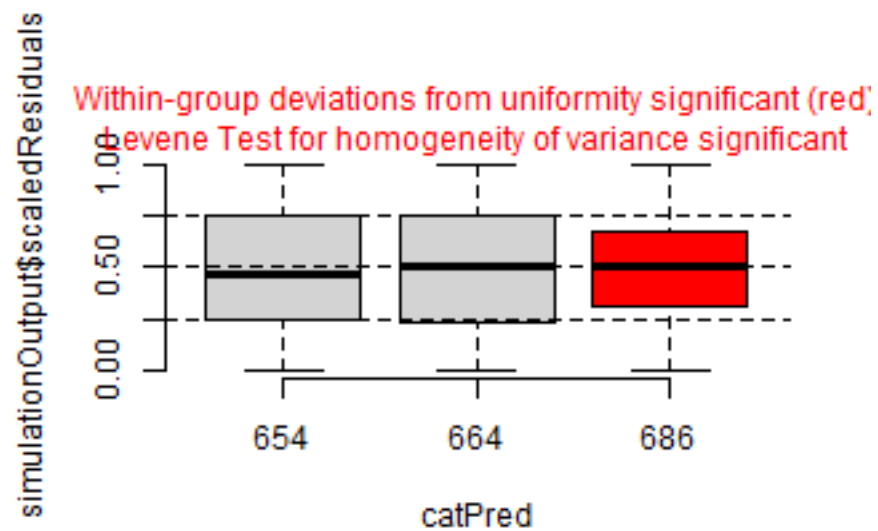
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



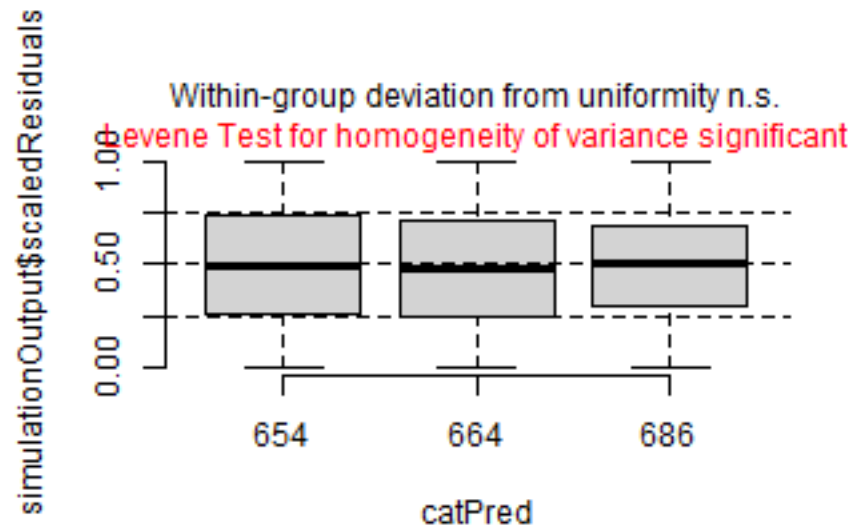
```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



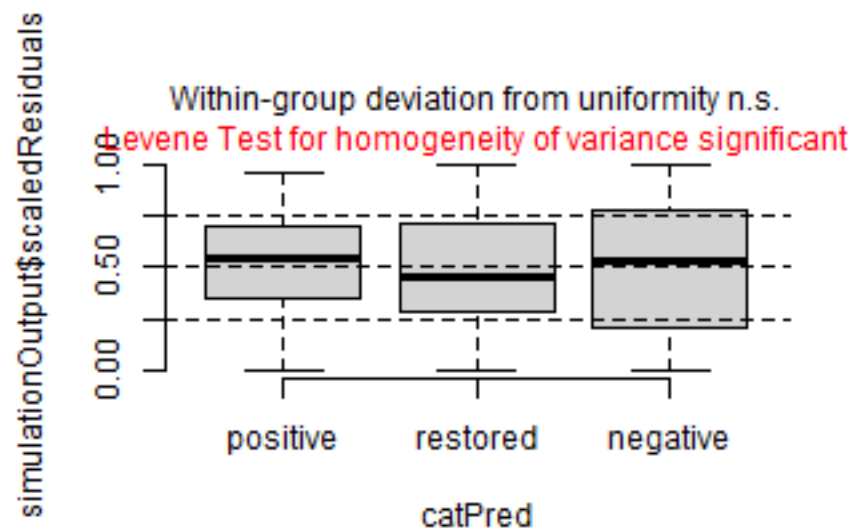
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



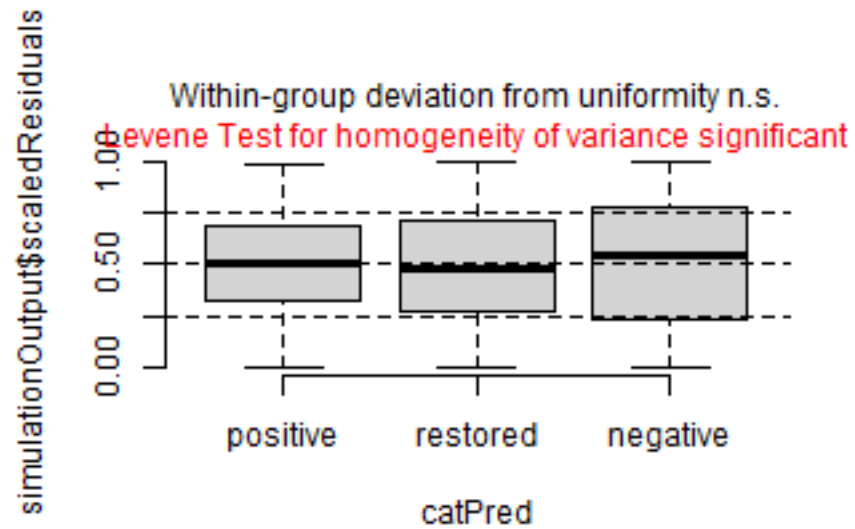
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



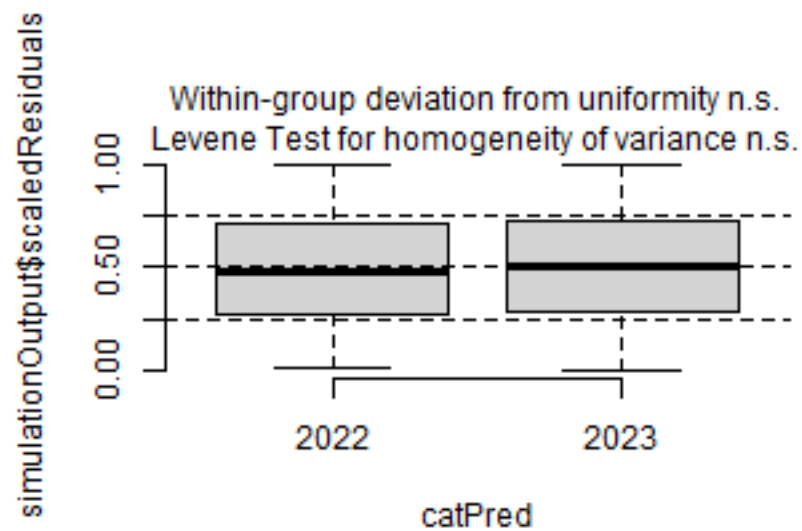
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



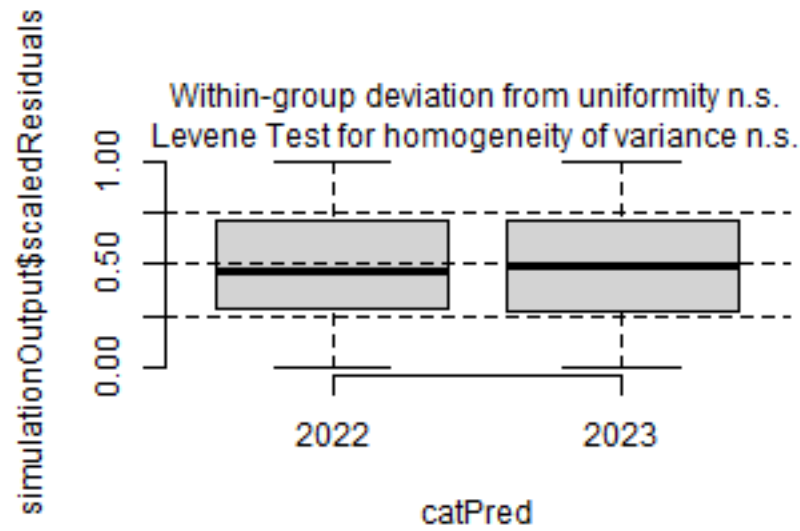
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



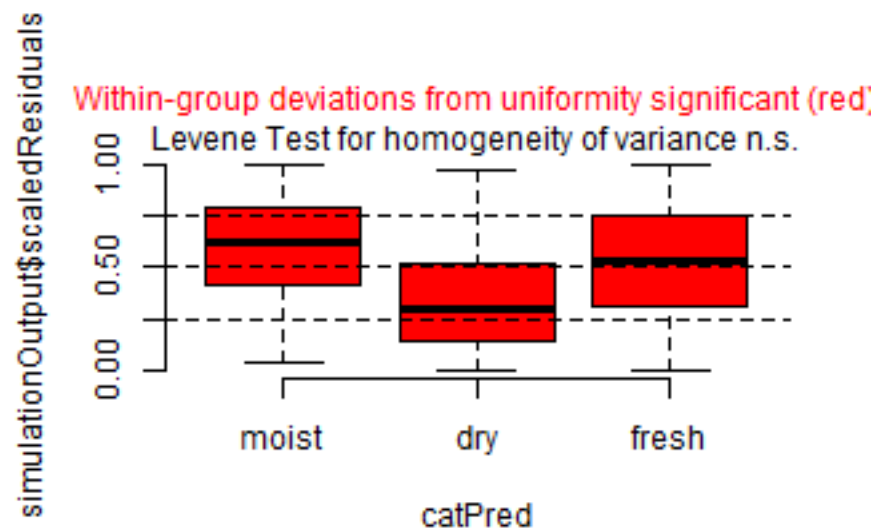
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



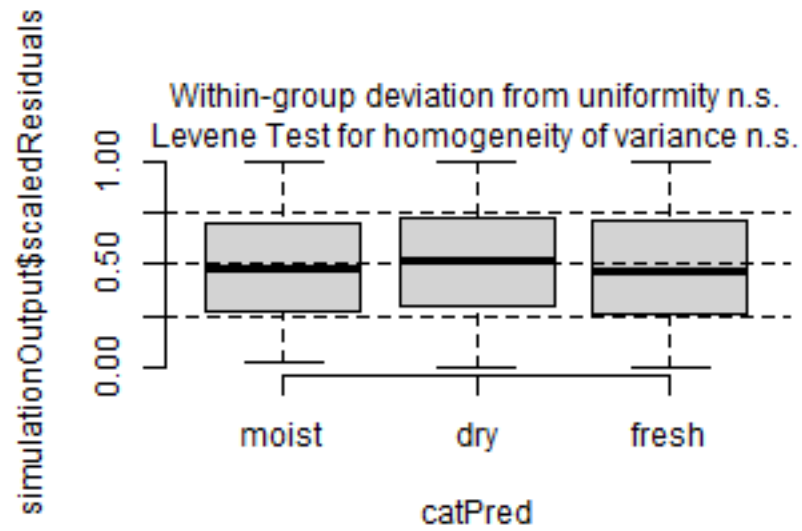
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



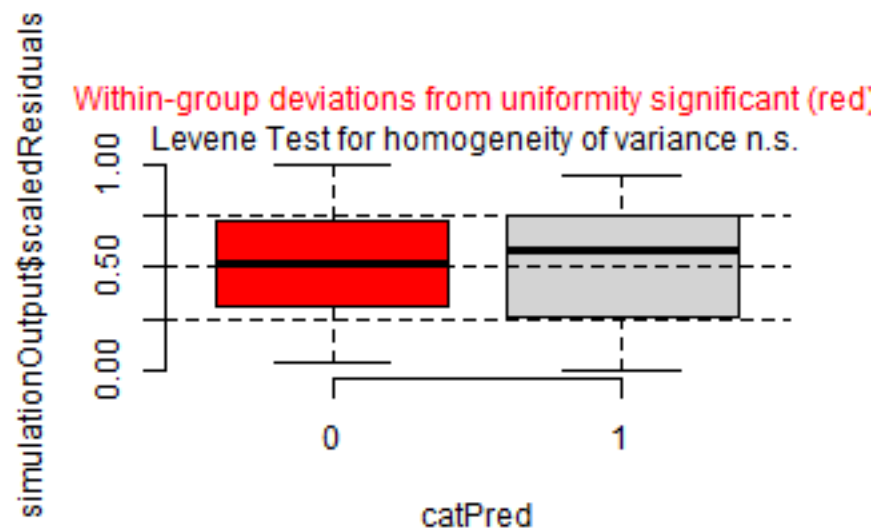
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



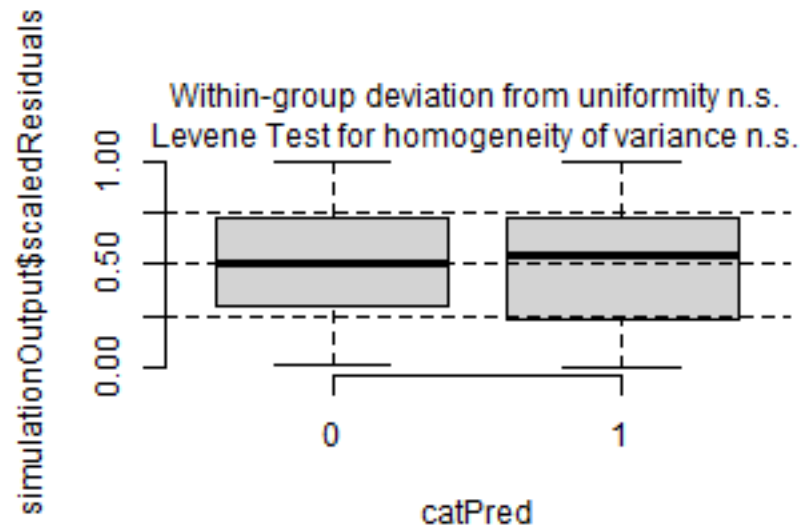
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



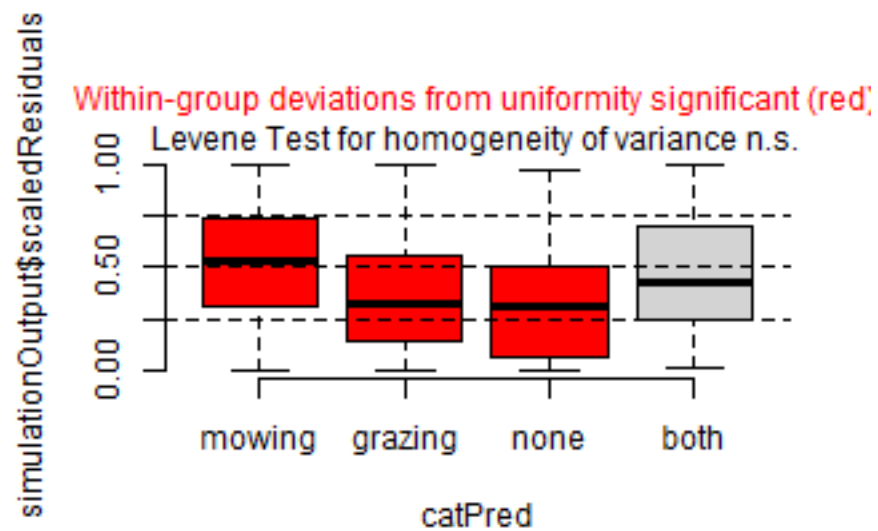
```
plotResiduals(simulation_output_1$scaledResiduals, sites$fertilized)
```



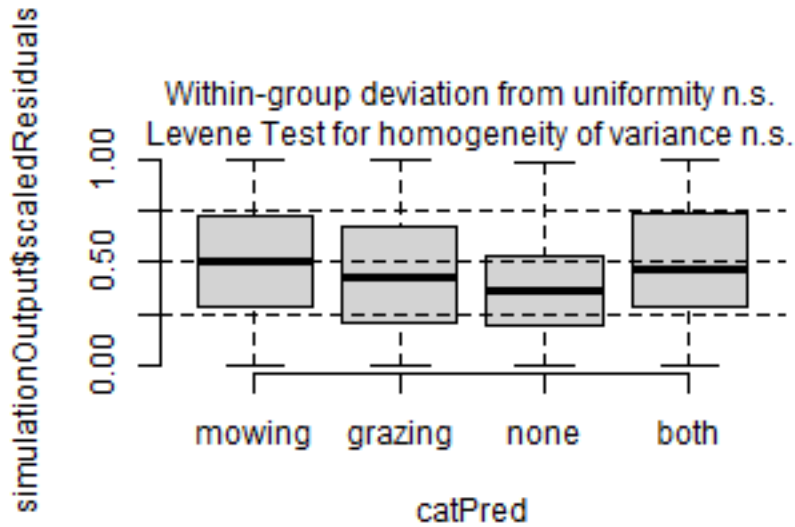
```
plotResiduals(simulation_output_2$scaledResiduals, sites$fertilized)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



## Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##              GVIF Df  GVIF^(1/(2*Df))
## esy4          3.487008 2      1.366511
## site.type     1.433677 2      1.094241
## eco.id        1.070092 2      1.017080
## obs.year      1.020821 1      1.010357
## esy4:site.type 4.545232 4      1.208355
```

```
car::vif(m_2)
```

```
##              GVIF Df  GVIF^(1/(2*Df))
## esy4          12.555214 2      1.882374
## site.type     1.596349 2      1.124040
## eco.id        1.775060 2      1.154259
## obs.year      1.032335 1      1.016039
## hydrology     1.341390 2      1.076190
## esy4:site.type 5.158284 4      1.227618
## esy4:eco.id   9.836240 3      1.463766
```

## Model comparison

### R2 values

```
MuMIn::r.squaredGLMM(m_1)
##           R2m           R2c
## [1,] 0.1519338 0.6740039
MuMIn::r.squaredGLMM(m_2)
##           R2m           R2c
## [1,] 0.3118902 0.6957894
```



## AICc

Use AICc and not AIC since ratio  $n/K < 40$  Burnham & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMin::AICc(m_1, m_2) %>%
  arrange(AICc)
##      df      AICc
## m_2 19 -1220.935
## m_1 14 -1190.718
```

## Predicted values

### Summary table

```
car::Anova(m_2, type = 3)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##              Chisq Df Pr(>Chisq)
## (Intercept)  454.2336  1 < 2.2e-16 ***
## esy4         12.9914  2   0.00151 **
## site.type     8.4332  2   0.01475 *
## eco.id        1.2761  2   0.52832
## obs.year      2.6442  1   0.10393
## hydrology     39.1188  2  3.202e-09 ***
## esy4:site.type  2.0074  4   0.73439
## esy4:eco.id    2.2107  3   0.52985
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_2)

## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: y ~ esy4 * (site.type + eco.id) + obs.year + hydrology + (1 |
##      id.site)
##      Data: sites
##
##           AIC          BIC      logLik -2*log(L)  df.resid
##    -1222.2    -1138.2      630.1    -1260.2      597
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2199 -0.5504  0.0267  0.4701  5.5476
##
## Random effects:
##  Groups   Name                Variance Std.Dev.
## id.site  (Intercept)  0.005939  0.07706
## Residual                    0.004706  0.06860
## Number of obs: 616, groups: id.site, 176
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)    0.422513   0.019824  21.313
## esy4R22         0.030833   0.017862   1.726
## esy4R1A        -0.070045   0.023888  -2.932
```

```
## site.type.L          0.044359  0.017162  2.585
## site.type.Q          0.013490  0.012528  1.077
## eco.id664           -0.020491  0.018736 -1.094
## eco.id686           -0.014544  0.018088 -0.804
## obs.year2023        -0.021419  0.013172 -1.626
## hydrologyfresh       0.094613  0.017296  5.470
## hydrology moist     0.113187  0.019863  5.698
## esy4R22:site.type.L -0.011640  0.022400 -0.520
## esy4R1A:site.type.L  0.034487  0.040824  0.845
## esy4R22:site.type.Q -0.015240  0.015333 -0.994
## esy4R1A:site.type.Q  0.001186  0.028152  0.042
## esy4R22:eco.id664   -0.027419  0.022264 -1.232
## esy4R22:eco.id686   -0.021702  0.020536 -1.057
## esy4R1A:eco.id686    0.016524  0.031184  0.530

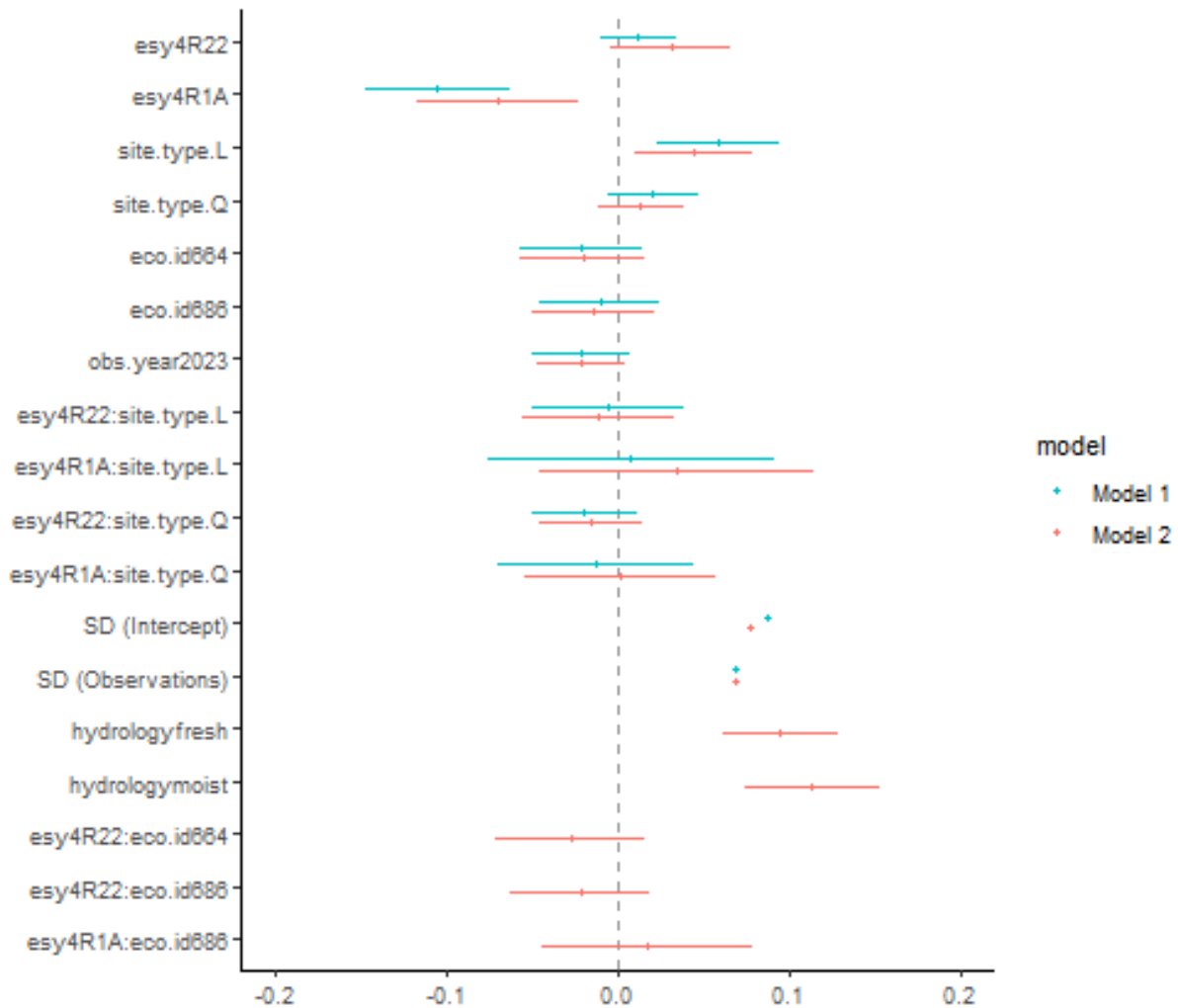
##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it

## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

### Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  xlim(-0.2, 0.2) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```



## Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_2,
  revpairwise ~ esy4,
  type = "response"
))
```

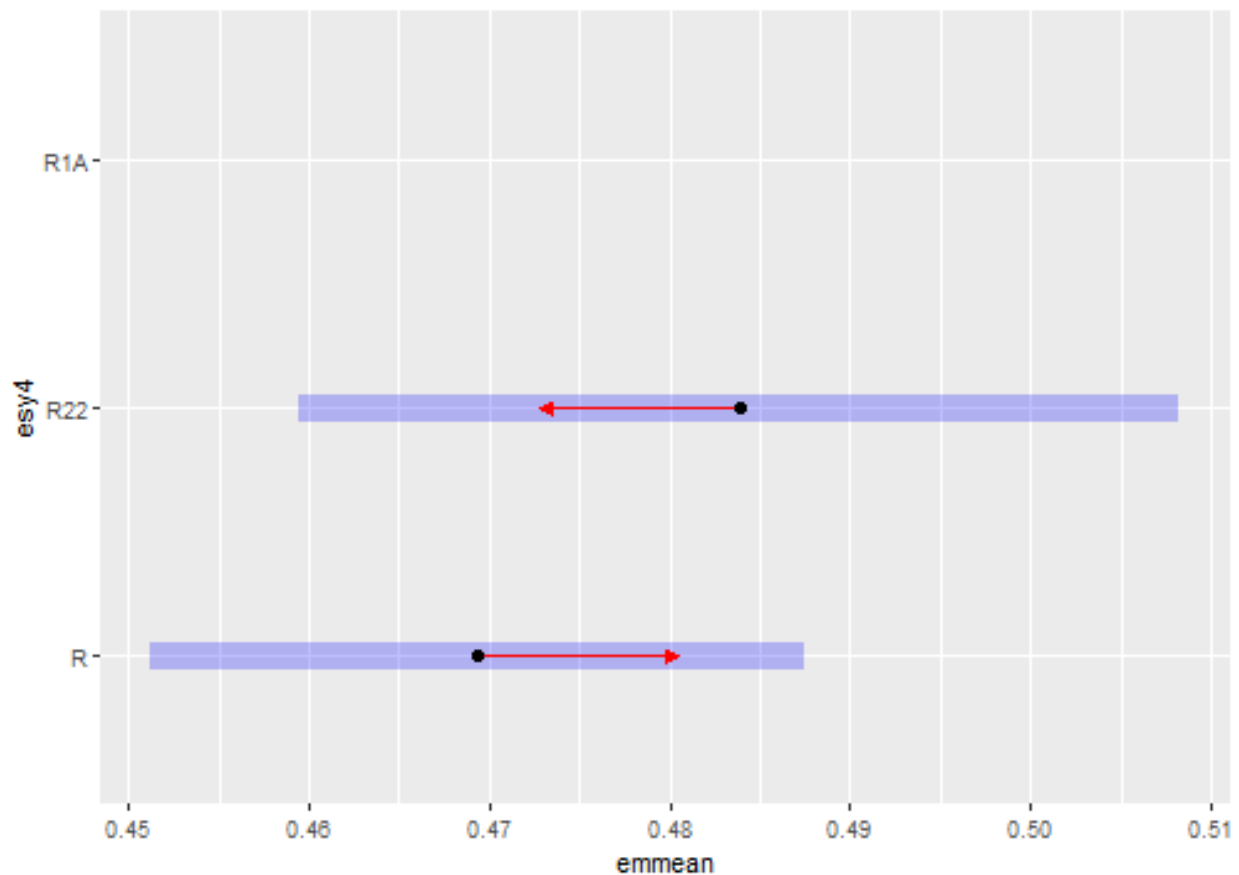
## ESY EUNIS Habitat type

```
## $emmeans
## esy4 emmean      SE    df lower.CL upper.CL
## R      0.469 0.00918 157      0.451      0.488
## R22    0.484 0.01240 299      0.459      0.508
## R1A nonEst      NA    NA         NA         NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
```

```
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##   contrast   estimate      SE  df t.ratio p.value
## R22 - R      0.0145 0.0113 580   1.275  0.2028
## R1A - R      nonEst    NA  NA      NA      NA
## R1A - R22    nonEst    NA  NA      NA      NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 2 estimates
```

```
plot(emm, comparison = TRUE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



```
(emm <- emmeans(
  m_2,
```

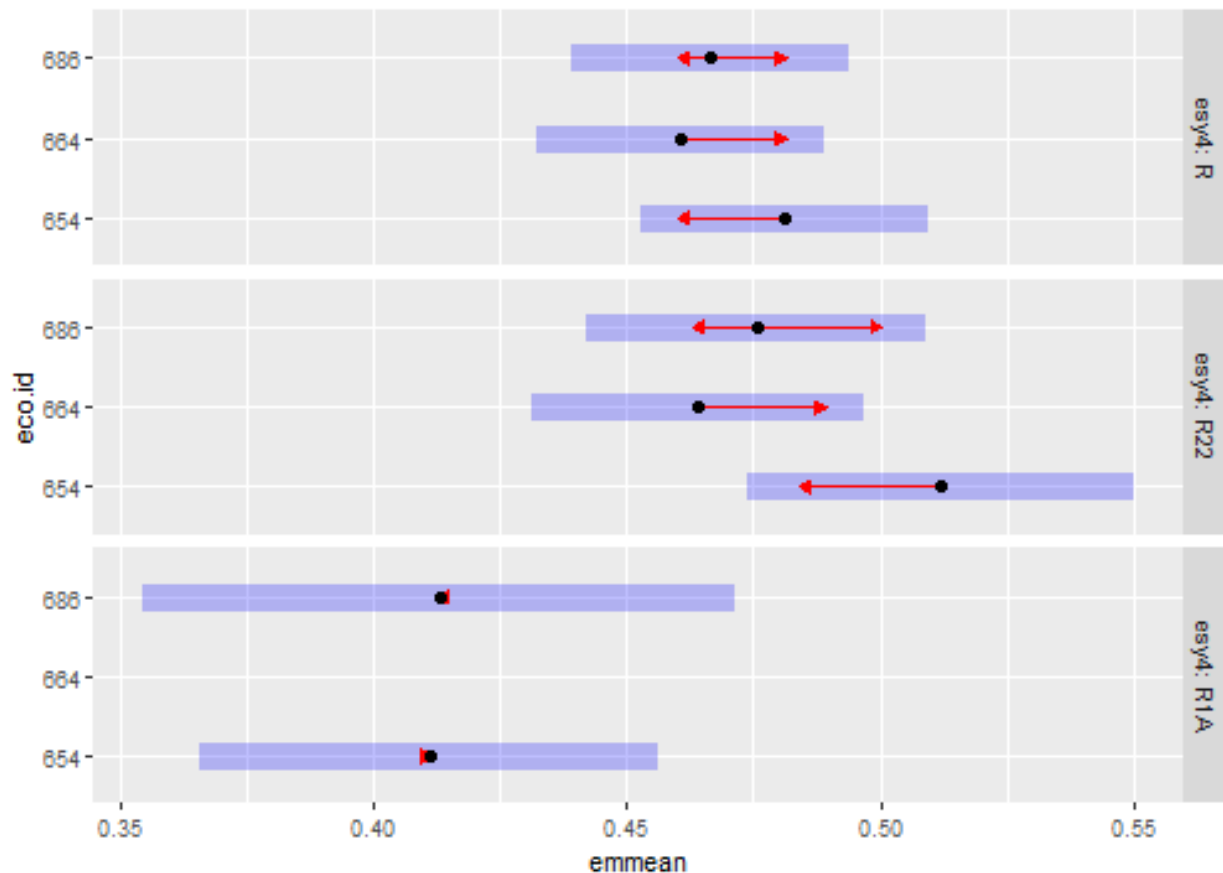
```
revpairwise ~ eco.id | esy4,
type = "response"
))
```

### Habitat type x Region

```
## $emmeans
## esy4 = R:
##   eco.id emmean      SE df lower.CL upper.CL
##   654      0.481 0.0144 204    0.453    0.510
##   664      0.461 0.0144 219    0.432    0.489
##   686      0.467 0.0139 185    0.439    0.494
##
## esy4 = R22:
##   eco.id emmean      SE df lower.CL upper.CL
##   654      0.512 0.0194 376    0.474    0.550
##   664      0.464 0.0166 311    0.431    0.497
##   686      0.476 0.0170 310    0.442    0.509
##
## esy4 = R1A:
##   eco.id emmean      SE df lower.CL upper.CL
##   654      0.411 0.0229 194    0.366    0.456
##   664    nonEst      NA  NA        NA        NA
##   686      0.413 0.0298 414    0.354    0.472
##
## Results are averaged over the levels of: site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## esy4 = R:
##   contrast                estimate      SE df t.ratio p.value
##   eco.id664 - eco.id654 -0.02049 0.0193 236  -1.063  0.5378
##   eco.id686 - eco.id654 -0.01454 0.0186 257  -0.782  0.7146
##   eco.id686 - eco.id664  0.00595 0.0187 251   0.318  0.9458
##
## esy4 = R22:
##   contrast                estimate      SE df t.ratio p.value
##   eco.id664 - eco.id654 -0.04791 0.0226 387  -2.118  0.0876
##   eco.id686 - eco.id654 -0.03625 0.0224 408  -1.619  0.2386
##   eco.id686 - eco.id664  0.01166 0.0207 335   0.565  0.8390
##
## esy4 = R1A:
##   contrast                estimate      SE df t.ratio p.value
##   eco.id664 - eco.id654    nonEst      NA  NA        NA        NA
##   eco.id686 - eco.id654  0.00198 0.0310 413   0.064  0.9491
##   eco.id686 - eco.id664    nonEst      NA  NA        NA        NA
##
## Results are averaged over the levels of: site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for varying family sizes
plot(emm, comparison = TRUE)
```

```
## Warning: Comparison discrepancy in group "R1A", eco.id654 - eco.id686:
```

```
## Target overlap = 0.9675, overlap on graph = -0.7438
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



```
(emm <- emmeans(
  m_2,
  revpairwise ~ site.type | esy4,
  type = "response"
))
```

### Habitat type x Site type

```
## $emmeans
## esy4 = R:
## site.type emmean      SE df lower.CL upper.CL
## positive  0.444 0.0184 262  0.407  0.480
## restored  0.458 0.0102 173  0.438  0.479
## negative  0.506 0.0170 127  0.473  0.540
##
```

```

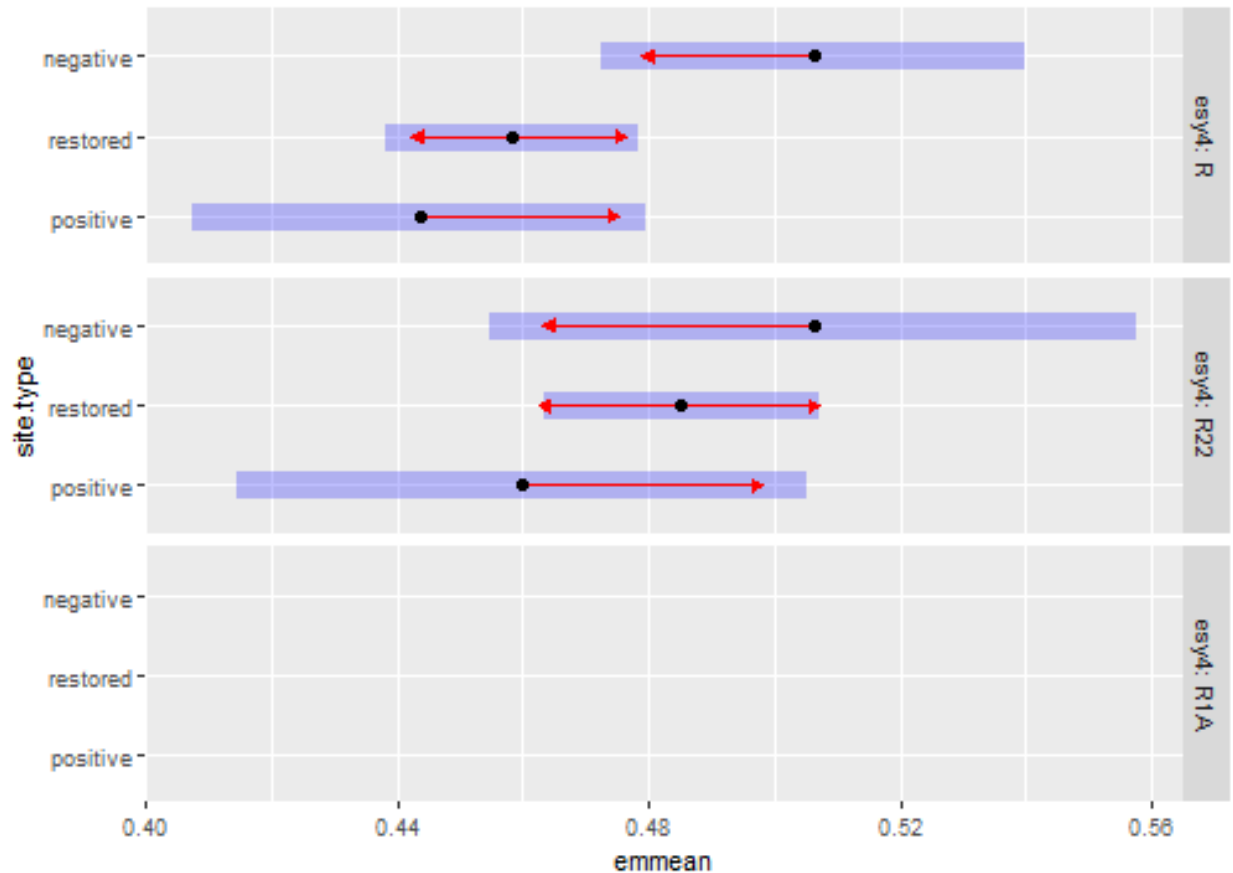
## esy4 = R22:
##   site.type emmean      SE  df lower.CL upper.CL
##   positive  0.460 0.0230 449    0.415    0.505
##   restored  0.485 0.0110 143    0.464    0.507
##   negative  0.506 0.0261 381    0.455    0.558
##
## esy4 = R1A:
##   site.type emmean      SE  df lower.CL upper.CL
##   positive nonEst    NA  NA      NA      NA
##   restored nonEst    NA  NA      NA      NA
##   negative nonEst    NA  NA      NA      NA
##
## Results are averaged over the levels of: eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## esy4 = R:
##   contrast      estimate      SE  df t.ratio p.value
##   restored - positive  0.0148 0.0210 223    0.706  0.7599
##   negative - positive  0.0627 0.0250 177    2.509  0.0346
##   negative - restored  0.0479 0.0192 198    2.491  0.0360
##
## esy4 = R22:
##   contrast      estimate      SE  df t.ratio p.value
##   restored - positive  0.0253 0.0254 339    0.996  0.5797
##   negative - positive  0.0463 0.0348 417    1.331  0.3786
##   negative - restored  0.0210 0.0277 472    0.758  0.7288
##
## esy4 = R1A:
##   contrast      estimate      SE  df t.ratio p.value
##   restored - positive  0.0378 0.0340 446    1.113  0.5068
##   negative - positive  0.1115 0.0579 390    1.925  0.1329
##   negative - restored  0.0737 0.0538 375    1.371  0.3574
##
## Results are averaged over the levels of: eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
plot(emm, comparison = TRUE)

## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).

```



## Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] emmeans_1.11.1  DHARMa_0.4.7    patchwork_1.3.0 ggbeeswarm_0.7.2
## [5] lubridate_1.9.4 forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4
## [9] purrr_1.0.4     readr_2.1.5     tidyr_1.3.1      tibble_3.2.1
```



```

## [13] ggplot2_3.5.2    tidyverse_2.0.0 here_1.0.1
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.4      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.1.6       magrittr_2.0.3     multcomp_1.4-28
## [7] compiler_4.5.0    mgcv_1.9-1         vctrs_0.6.5
## [10] pkgconfig_2.0.3   crayon_1.5.3       fastmap_1.2.0
## [13] backports_1.5.0   labeling_0.4.3     utf8_1.2.5
## [16] ggstance_0.3.7    promises_1.3.2     rmarkdown_2.29
## [19] tzdb_0.5.0        nloptr_2.2.1       bit_4.6.0
## [22] xfun_0.52         later_1.4.2        broom_1.0.8
## [25] parallel_4.5.0    R6_2.6.1           gap.datasets_0.0.6
## [28] stringi_1.8.7     qgam_2.0.0         RColorBrewer_1.1-3
## [31] car_3.1-3         boot_1.3-31        estimability_1.5.1
## [34] Rcpp_1.0.14       iterators_1.0.14   knitr_1.50
## [37] zoo_1.8-14        parameters_0.25.0  httpuv_1.6.16
## [40] Matrix_1.7-3      splines_4.5.0      timechange_0.3.0
## [43] tidysselect_1.2.1 rstudioapi_0.17.1  abind_1.4-8
## [46] yaml_2.3.10       MuMIn_1.48.11      doParallel_1.0.17
## [49] codetools_0.2-20  lattice_0.22-6     plyr_1.8.9
## [52] bayestestR_0.15.3 shiny_1.10.0        withr_3.0.2
## [55] evaluate_1.0.3    marginaleffects_0.25.1 survival_3.8-3
## [58] pillar_1.10.2     gap_1.6            carData_3.0-5
## [61] foreach_1.5.2     stats4_4.5.0       reformulas_0.4.1
## [64] insight_1.2.0     generics_0.1.4     vroom_1.6.5
## [67] rprojroot_2.0.4   hms_1.1.3          scales_1.4.0
## [70] minqa_1.2.8       xtable_1.8-4       glue_1.8.0
## [73] tools_4.5.0       data.table_1.17.2  lme4_1.1-37
## [76] mvtnorm_1.3-3     grid_4.5.0         rbibutils_2.3
## [79] datawizard_1.1.0  nlme_3.1-168       Rmisc_1.5.1
## [82] performance_0.13.0 beeswarm_0.4.0     vipor_0.4.7
## [85] Formula_1.2-5     cli_3.6.5          gtable_0.3.6
## [88] digest_0.6.37     pbkrtest_0.5.4     TH.data_1.1-3
## [91] farver_2.1.2      htmltools_0.5.8.1  lifecycle_1.0.4
## [94] mime_0.13         bit64_4.6.0-1      dotwhisker_0.8.4
## [97] MASS_7.3-65

```